

Specialisation:

Stochastic Modelling, Machine Learning and Wireless Networks.

EDUCATION

Télécom Paris (LTCI lab) - INRIA (MathNet team)

Mar 2025 - Mar 2028

Ph.D. in Applied Probability

Paris, France

- Subject: Stochastic Modelling of Non-Terrestrial Networks (NTNs).
- Areas: Stochastic geometry, ergodic theory, Palm calculus.
- Side activity: Part of ESA's NTN forum for TN-NTN integration and standards definition.
- Supervisors: Prof. F. Baccelli (ENS Paris), Prof. L. Decreusefond (Télécom Paris), Dr. N. Soprano-Loto (Inria).

École Polytechnique Fédérale de Lausanne (EPFL)

Apr 2024 - Sept 2024

Master's Thesis Visiting Student - Computer Science

Lausanne, Switzerland

- Thesis: Dynamic Bayesian Optimisation (DBO) for improving the performance of Cellular Networks.
- Key Results: Theorised a novel probabilistic mobility model and developed a cellular network simulator to evaluate DBO algorithm for downlink power optimisation against naïve approaches.
- Supervisors: Prof. P. Thiran (EPFL), Dr. A. Bardou (EPFL).

École Polytechnique

Sept 2022 - Sept 2024

MSc in Computer Science - Track: Internet of Things

Paris, France

- Double Major: Machine Learning, Computer Networks; GPA: 3.89/4.
- Key subjects: Machine, Reinforcement and Deep Learning, Federated Learning, Large scale optimization, Computer and Wireless Networks, Embedded Systems.

Linköping University (LIU)

Aug 2021 - Jan 2022

Exchange Programme - Industrial Engineering

Linköping, Sweden

- GPA: 4.5/5.

Alma Mater Studiorum – Università di Bologna (UNIBO)

Sept 2019 - Jul 2022

BSc in Industrial Engineering - Minor: Information & Communication Technology

Bologna, Italy

- Final mark: 110/110 cum LAUDE; GPA: 28.94/30.
- Thesis: Conception and Design of a Predictive Maintenance System for the Automotive Sector.
- Key subjects: Signal Processing, Operation Research, Electronics, Logistics, Stochastic Process Control.

WORK EXPERIENCE

INRIA (MATHNET team)

Sept 2024 - Mar 2025

R&D Engineer - Predoctoral

Paris, France

- Research as part of the EU-funded INSTINCT project, applying stochastic geometry and filtering theory for modelling Joint Communication and Sensing (JCAS) scenarios in 6G networks.

- Pursued academic training on advanced probability theory and satellite communications.
- Supervisor: Prof. F. Baccelli (ENS Paris).

Cisco Meraki

Apr 2023 - Sept 2023

Software Engineer Intern - Data Science team

Paris, France

- Theorised and developed a novel time series prediction model and a causal inference algorithm. Both algorithms were used to build an anomaly detection product for L2 devices.
- Key Results: Conference publication [1].
- Supervisors: Prof. T. H. Clausen (École Polytechnique) and Dr. Z. Yao (Cisco Meraki).

National Inter-University Consortium for Telecommunications (CNIT)

Nov 2022 - Apr 2023

Scientific Research Partner

Bologna, Italy

- Developed a low computational cost algorithm for RIS-aided UAV tracking in a non-line-of-sight scenario.
- Key Results: Contribution to publication [2].
- Supervisor: Prof. F. Guidi (UNIBO).

Cluster Reply

Mar 2022 - May 2022

Junior IT Consultant Intern

Turin, Italy

- Developed a predictive maintenance algorithm for vehicles failure prediction, combining statistical process control and regression-based techniques.
- Jointly led sales and product development.
- Key Results: Algorithm got acquired by third-part company (Movyon) for Tunnel Fault Monitoring.
- Supervisors: Prof. F. Guidi (UNIBO) and S. Agostini, Associate Partner (Cluster Reply).

Junior Enterprise Bologna (JEBO)

Jul 2021 - Jul 2022

Vice President

Bologna, Italy

- Managed 90 human resources, overseeing 51K € in revenue (+60%).
- Led partnership management – created 10 new collaborations.
- Creator of “Coffee with Entrepreneurs” career orientation program (moderator of 3 events):
 - 30 entrepreneurs involved, 1K podcast views, avg. 30 attendees per event.

PUBLICATIONS

- [1] E. Mengoli, Z. Yao, W. Wei, and T. Clausen, “Develop end-to-end anomaly detection system,” in *2023 IEEE International Conference on Data Mining Workshops (ICDMW)*, IEEE, 2023, pp. 1370–1379.
- [2] F. Guidi, A. Guerra, E. Mengoli, and A. Zanella, “A statistical characterization of communication performance in ris-aided networks,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2024, pp. 12 921–12 925.
- [3] E. Mengoli, L. Moll, V. Strozzi, and E.-M. El-Mhamdi, “On the byzantine fault tolerance of signsgd with majority vote,” *arXiv preprint arXiv:2502.19170*, 2025.

INVITED TALKS

13ème Atelier en Évaluation des Performances

Dec, 2024
Toulouse, France

CONTRIBUTED TALKS

EuCNC & 6G Summit <i>Demo booth presentation.</i>	June, 2025 <i>Poznań, Poland</i>
5th Italian Meeting on Probability and Mathematical Statistics <i>Speaker and organizer of a contributed session.</i>	June, 2026 <i>Palermo, Italy</i>
Stochastic Networks Conference <i>Poster session.</i>	June, 2026 <i>Chicago, USA</i>

RESEARCH SCHOOLS

Random walks: applications and interactions	Jan, 2026 <i>CIRM, Marseille, France</i>
XXIX Brazilian School of Probability <i>Poster session.</i>	August, 2026 <i>Rio de Janeiro, Brazil</i>

EXTRACURRICULAR ACTIVITY

Federazione Italiana Pallacanestro (FIP) <i>Basketball referee</i>	Dec 2014 - Sept 2022 <i>Italy</i>
▫ “Trofeo delle regioni” (2019) - selected as one of the two best referees in Tuscany. ▫ “14th World Maxibasketball Fimba Championship” (2017).	

SKILLS SUMMARY

- **Languages:** Italian: Native, English: Proficient, Spanish: Fluent, French: Fluent.
- **Tools:** Python, SQL, Java, C, LaTeX, AWS, Azure, Docker, Git, GitLab, GitHub, MySQL, Arduino, Trello, Jira, Microsoft Office Suite, Google Suite, Google Data Studio.